# Sihan Qin

Email | LinkedIn | Google Scholar | GitHub | Homepage | Guangzhou, P.R. China

## EDUCATION

**Sun Yat-sen University** Guangzhou, China

***B.E. in Computer Science and Technology (Artificial Intelligence & Big Data)*** Jun. 2027(Expected)

***GPA: 3.8/4.0***

- **Core Courses:** Machine Learning & Data Mining, Introduction to Robotics, Modern Control Systems, Principles & Applications of Multimodal Deep Models, Artificial Intelligence, Optimization Theory, Pattern Recognition
- **Awards:** 3rd-Class University Scholarship (2023, 2024); Honorable Mention, Mathematical Contest in Modeling 2025

## RESEARCH EXPERIENCE

**RL Post-Training for VLA Models in Multi-task Settings** Shenzhen, China

*ISEE Lab, Sun Yat-sen University, advised by Prof. Weishi Zheng* Apr. 2026 – Sep.2026

- Trained and evaluated VLA models (StarVLA & $\boldsymbol{\pi_{0.5}}$) with RL Token in simulated environments, including RoboTwin2.0 and LIBERO.
- Building a real-world end-to-end $\boldsymbol{\pi_{0.5}}$(with RL Token) pipeline for AgileX NERO/Piper arms, spanning model deployment, teleoperated data collection, training, and online inference.

**Physical Autoregressive Model for Robotic Manipulation without Action Pretraining** Guangzhou, China

*HCP Lab, Sun Yat-sen University, advised by Prof. Guangrun Wang and Prof. Liang Lin* Jul. 2025 – Mar. 2026

- Proposed and developed a physical autoregressive model that transfers physical priors from video pretraining to robotic manipulation, enabling video-consistent action generation without action pretraining.
- Conducted training and evaluation on ManiSkill, LIBERO, and CALVIN; tuned key hyperparameters including batch size, learning rate, and LR scheduler, improving simulation success rates from about 30% to over 90%.
- Conducted real-robot experiments on a Franka Panda arm and collected 80 teleoperated trajectories for supervised learning, improving model generalization and real-world robustness; also contributed to paper writing.

**Procedural 3D Image Generation from Lung CT Scans** Guangzhou, China

*Robotic and Intellligence Computing Lab, Sun Yat-sen University, advised by Prof. Kai Huang* May – Sep. 2024

- Curated 20 full respiratory-cycle lung CT samples from public datasets and performed threshold segmentation, meshing, and point cloud conversion with scikit-image and Open3D.
- Separated bronchi and lung boundaries using RANSAC, curvature clustering, Alpha Shape, and voxel filtering to support downstream lesion analysis.

## PUBLICATION

Zijian Song, **Sihan Qin**, Tianshui Chen, Liang Lin, Guangrun Wang, *Physical Autoregressive Model for Robotic Manipulation without Action Pretraining,* arXiv:2508.09822.

Zijian Song, Qichang Li, **Sihan Qin**, Yuhao Chen, Tianshui Chen, Liang Lin, Guangrun Wang, *Learning Physics from*

*Pretrained Video Models: A Multimodal Continuous and Sequential World Interaction Models for Robotic Manipulation*, arXiv:2603.00110, **icmr 2026**.

Zijian Song, Xiaoxin Lin, Qiuming Huang, **Sihan Qin**, Guangrun Wang, Liang Lin, *SIRI-Bench: Challenging VLMs' Spatial Intelligence through Complex Reasoning Tasks,* arXiv:2506.14512.

## INTERNSHIP EXPERIENCE

**BrainCo Technology Co., Ltd.** Shenzhen, China

*Embodied AI Algorithm Intern* Sep.2026 – Present

- Built an end-to-end data collection platform for robotic prosthetics, integrating a MANUS teleoperation glove, a BrainCo Revo3 dexterous hand, a D405 camera, and an electromyography (EMG) armband.
- Implemented hybrid motion retargeting with GeoRT for the thumb and AnyTeleop for the other four fingers, and established time synchronization across teleoperation, visual, and EMG data streams.
- Supporting a vision-language-action (VLA) + EMG prosthetics project aimed at preserving user control through EMG-based action selection and enabling dexterous manipulation across objects and scenes through VLA generalization.

**Guangzhou Shuzhi Technology** Guangzhou, China

*Algorithm Intern* Sep. – Oct. 2025

- Contributed to prompt engineering for an academic–industry matching platform.
- Diagnosed key flaws in legacy AI interactions and implemented a prompt framework combining UltraRAG + Deep Research with multi-round retrieval-LLM for deep analysis of project value, team structure, and research outcomes.
- Structured 5 key LLM-generated analytical results into JSON and stored them in databases to support front-end development.

## OTHERS

**Zotero Paper Partner** Guangzhou, China

*Independent Project* Apr. 2026

- Developed an attention-friendly Zotero plugin that answers questions inside native notes.

**Mathematical Contest in Modeling 2025** Guangzhou, China

*Olympic Medal Table Prediction, Honorable Mention* Feb. 2025

- Built regression, random forest, and SVM models, combined with SMOTE, Bayesian optimization, and Monte Carlo simulation, to predict medal counts and analyze key influencing factors.

## CAMPUS LEADERSHIP

Sun Yat-sen University Student Union & School of Computer Science Student Union Sep. 2023 – Jul. 2024

*General Affairs Department Member; "Red Kapok" Event Coordinator*

## SKILLS

**Technical:** Machine Learning, Deep Learning, Robotics, PyTorch, Linux, EMG

- **Real Robot**: Franka, Agile-X(Nero/Piper), so-101 and revo3(21DoF Dexterous Hand)
- **Simulation Platform**: ManiSkill, LIBERO and RoboTwin2.0
- **VLA model**: $\pi_{0.5}$, StarVLA, smolVLA, ACT